

Methodological Falsifiability and Dependency Integrity Standard v2.3

v2.3 Current Release Integration

Release: MathGov Core Release 2026.09 v12.6 / SGP v8.5 - Independent Open-Source Qualification-Continuity Release

Exact current companion pins: Canon v12.6; SGP v8.5; ripple.md v5.5; Agent System v12.5; CSV v2.4; Cascade v2.6; Reproducibility v1.4; WDBIP v1.6; RLS Validation v2.6; Primer v4.4; Public Introduction v12.6; PC-AEP/MFDI/Source-Coupling v2.3; Aligners Sheet v5.6.

Integrity surface	Current requirement	Claim boundary
Claim typing	Material empirical, causal, formal, operational, legal-regulatory, normative, and metaphysical-horizon claims declare their evidence and authority boundaries.	The protocol does not make every legal or normative commitment physically falsifiable.
Warrant-domain bridge	A material conclusion identifies the domain of the offered warrant and any bridge required when the conclusion crosses scientific, engineering, legal-regulatory, institutional, or normative domains.	Authority, compliance, certification, performance, and rhetoric cannot silently change claim type.
Dependency integrity	Starting assumptions, definitions, models, thresholds, and inherited sources are versioned; material changes trigger re-derivation.	Silent retrofitting is prohibited.
Revision trigger	A claim identifies evidence or conditions that would force revision, narrowing, rerun, or retirement.	Fluency and coherence are not external confirmation.
Capability claim integrity	A material capability term is operationally defined and bound to observable criteria, mechanism or system composition, evidence, validity conditions, alternatives, a falsifier, evidence maturity, permitted wording, and explicit nonclaims.	Neither component reduction nor anthropomorphic language settles the capability claim.

Source-boundary rule: If this compact standard conflicts with the RippleLogic Canon, the Canon controls.

v2.3 Frozen Evaluation Binding and Milestone Claim Discipline

Before consequential evaluation, the run freezes and versions the decision question, system boundary, configuration, evidence schema, stakeholder scope, rights interpretation, scenario families, thresholds, uncertainty treatment, weighting protocol, control requirements, stopping conditions, and authorization pathway. A material post-result alteration creates a new run or declared revision branch.

Implementation milestones are classified narrowly. Successful formal derivation, compilation, synthesis, simulation, hardware implementation, or integration may defeat a specific impossibility hypothesis, but it does not prove empirical superiority, domain validity, physical safety, ontology, rights compatibility, or deployment readiness. Claims must name the exact falsification criterion tested and the alternatives that remain open.

Purpose

This compact standard defines the v12-line methodological integrity discipline for MathGov and RippleLogic, introduced at v12.0 and retained in the current v12.6 release. It adopts the bounded principle that claim-bearing elements must not survive merely because they are plausible, fluent, approved, compliant, or locally useful. A material claim must declare what kind of claim it is, what it depends on, what would force revision, what alternatives remain open, and what downstream elements must be rerun if the foundation changes.

It is not a sixth gate. It does not replace Reality Grounding, RF/NCRC, TRC, CSV, SGP, RLS, PC-AEP, domain science, or external validation.

Adopted principles and bounded non-claims

Item	MathGov integration
Adopted principle	Material claims need external contact, falsification or revision triggers, explicit dependencies, and no hidden assumption authority.
Adopted principle	If a starting assumption, definition, threshold, operator, or model basis changes, the affected downstream chain must be rerun or claim strength must be narrowed.
Bounded non-claim	Not every normative definition is physically falsifiable. Normative commitments are audited for coherence, rights compatibility, stakeholder challenge, and application integrity.
Bounded non-claim	Probabilistic evidence is not invalid. It is required where uncertainty is real, but uncertainty must be disclosed and cannot be laundered into certainty.
Bounded non-claim	Governance may act proportionately under serious uncertainty to protect rights, bound ruin, require evidence, authorize bounded testing, or impose controls. Such action does not prove that the uncertain capability exists or that uncertainty has been reduced.
Bounded non-claim	Model revision is not forbidden. Silent retrofitting is forbidden. Revision must be versioned, disclosed, and rechecked.
Bounded non-claim	System-level operational capabilities may emerge from coordinated probabilistic, deterministic, memory, planning, feedback, tool, and human-control components. Component decomposition does not by itself refute the bounded capability, and the bounded capability does not establish consciousness, moral agency, generality, or authority.

Required Methodological Integrity Record

Field	Required content
claim_or_component	The claim, definition, threshold, operator, indicator, model, SGP interpretation, gate decision, validation result, or public release claim under review.
claim_type	empirical, causal, formal, operational, legal_regulatory, normative, or metaphysical_horizon. The display labels “legal-regulatory” and “metaphysical-horizon” MAY be used in prose; validator-facing records MUST use legal_regulatory and metaphysical_horizon.
claim_domain	The primary authority domain in which the conclusion is asserted: scientific_empirical, engineering_operational, formal_mathematical, legal_regulatory, normative_ethical, institutional_governance, commercial_descriptive, anthropomorphic_social, or metaphysical_horizon. A materially mixed conclusion SHOULD be split into separately typed claims.
warrant_domains	One or more domains from which support derives. Multiple domains MAY support one claim, but each material cross-domain relationship requires its own bridge record.
cross_domain_bridges	Zero or more bridge records. Each record contains from_warrant_domain, to_claim_domain, bridge_status, bridge_proposition, bridge_evidence_or_rationale, uncertainty, validity_or_jurisdiction_boundary, and revision_or_refusal_condition.
dependency_position	Where the claim sits in the cascade, standard, SGP flow, workbook, validation protocol, or public release surface.
starting_assumptions	Assumptions, definitions, priors, sources, thresholds, scope boundaries, or model premises that carry the claim.
definition_or_operator_used	The material definition, operator, threshold, indicator, or procedure being applied.
necessity_or_alternative_check	Whether necessity is claimed, what alternatives remain plausible, and why the stronger claim is or is not warranted.
evidence_or_test_surface	The observation, measurement, replication, derivation, simulation, formal proof, expert review, stakeholder challenge, or audit surface used.
falsification_or_revision_trigger	What result, evidence, challenge, failure, or domain change would force revision, narrowing, escalation, or refusal.
uncertainty_or_confidence_method	How uncertainty, confidence, model limits, disagreement, or unresolved evidence is

Field	Required content
	represented.
downstream_dependencies	Gates, scores, SGP outputs, claims, controls, authority actions, or release statements affected if this element changes.
re_derivation_scope_if_changed	Local patch, threshold update, gate rerun, TRC reopening, CSV rerun, SGP rerun, RLS recalculation, release versioning, or full refusal.
reviewer_status	reviewer-visible, contested, independently checked, domain-expert-warranted, or pending.
required_claim_action	proceed, narrow, control, redesign, escalate, or refuse.
operational_definition (conditional)	The measurable functional meaning of the capability term in the pinned configuration and domain.
observable_criteria (conditional)	The observations that would count as demonstration, non-demonstration, or failure of the capability.
mechanism_or_system_composition (conditional)	The components and interactions that instantiate the claimed capability without assuming any single component supplies the whole.
evidence_surface (conditional)	The logs, trials, benchmarks, traces, inspections, or other evidence that support the bounded claim.
validity_conditions (conditional)	Configuration, domain, environment, time, operator, and control conditions inside which the claim is licensed.
alternative_explanations (conditional)	Simpler or rival explanations such as scripting, operator intervention, memorization, benchmark fit, or interface effects.
falsification_or_revision_trigger (conditional)	The observation or boundary violation that would narrow, revise, suspend, or refute the claim.
evidence_maturity_stage (conditional)	Definition/derivation, computational realization, implementation conformance, projection/calibration lock, controlled empirical contact, operational qualification, or outcome/requalification evidence.
permitted_claim (conditional)	The strongest wording licensed by the present evidence and maturity stage.
explicit_nonclaims (conditional)	Stronger inferences not licensed, including consciousness, personhood, intrinsic goals, moral agency, generality, unrestricted autonomy, safety, legality, or authority where applicable.

Dependency-localized failure propagation

If a premise, primitive, definition, operator, evidence source, control, or inherited dependency fails, every materially dependent gate result, score, SGP output, authorization, validation claim, or public statement must be reopened, narrowed, suspended, rerun, or withdrawn. The record must identify the dependency path and re-derivation scope.

A component with an independent evidence and derivation path does not fail merely because another part of the framework fails. This rule prevents both monolithic immunity—protecting dependent claims after foundation failure—and indiscriminate whole-framework rejection—discarding independently supported components without testing their actual dependencies.

Status values

Status	Meaning	Default action
METHOD_SUPPORTED	Evidence, dependency, and revision conditions support the declared claim boundary.	Proceed within boundary.
ASSUMPTION_BOUND	Claim depends on visible assumptions that could affect strength or gate status.	Narrow or monitor.
ALT_EXPLANATION_OPEN	Plausible alternatives can explain the same result.	Downgrade necessity claim.
TEST_REQUIRED	Stronger evidence, derivation, replication, simulation, or review is needed.	Escalate before strong claim.
CONTESTED	Evidence, interpretation, stakeholder challenge, or expert judgment materially disputes the claim.	Escalate, narrow, or redesign.
REVISION_REQUIRED	A starting assumption, definition, threshold, operator, or model basis has failed or changed.	Rerun affected chain.
REFUSE_STRONGER_CLAIM	The requested public, deployment, validation, deterministic-selection, or safety claim exceeds support.	Refuse the stronger claim.

Claim-type rules

Claim type	Evidence discipline
Empirical	Observation, measurement, replication, current source, or empirical warrant.
Causal	Mechanism, intervention, counterfactual, domain model, or qualified causal warrant.
Formal	Explicit assumptions, definitions, derivation, proof, or formal verification.
Operational	Reproducible procedure, audit trail, role assignment, and reviewable outputs.
Legal-regulatory	Current authority source, jurisdiction, applicability boundary, interpretation status,

Claim type	Evidence discipline
	and contest or review path.
Normative	Declared ethical commitment, rights compatibility, consistency, stakeholder challenge, and auditability of application.
Metaphysical-horizon	Humility and meaning orientation only; no Tier 1-3 evidence, gate override, or authority by itself.

Claim-domain and warrant-domain non-substitution

A material claim **MUST** distinguish its `claim_type`, its primary `claim_domain`, and every material `warrant_domain` offered in support. Scientific or empirical evidence can support bounded descriptive or causal claims but does not by itself create legal authority or ethical legitimacy. Engineering performance can support bounded operational capability but does not by itself establish rights compatibility, lawful permission, or safety outside the tested envelope. Legal, regulatory, institutional, certification, compliance, or policy authority can govern permission, procedure, responsibility, testing, or review but does not by itself establish empirical truth, causal validity, technical capability, physical safety, or reduced epistemic uncertainty. Commercial or anthropomorphic terminology creates no stronger evidence or authority by itself.

When a conclusion draws on more than one warrant domain, the record **MUST** list all material warrant domains. For every warrant domain that differs from the claim domain, it **MUST** create a separate bridge record stating the bridge proposition, evidence or rationale, uncertainty, relevant validity or jurisdiction boundary, and the condition that would narrow, revise, or refuse the bridge. `BRIDGE_PARTIAL` permits only the supported narrower claim. `BRIDGE_UNSUPPORTED` requires narrowing, escalation, redesign, or refusal of the cross-domain conclusion and triggers `CROSS_DOMAIN_WARRANT_SUBSTITUTION` if the stronger claim is nevertheless asserted.

Non-retrofitting rule

A failed test, falsified prediction, changed foundation, or challenged dependency must not be hidden by silent tuning. The run must version the change, record the affected dependencies, and rerun every materially dependent gate, SGP interpretation, calculation, authority action, or public claim.

Proportionality rule

Methodological rigor is proportional to stakes. Tier 1 and low-stakes Tier 2 uses may use lighter records. Tier 3, public conformance, institutional deployment, high-stakes SGP use, validation claims, and physical/causal actions require stronger records.

Physical-admissibility claim discipline

A claim that a physical execution is safe, admissible, stable, reliable, or deployment-ready is a material claim and must name its external evidence surface. Governance permission, compliance status, documentation, certification, monitoring, and authority are not sufficient evidence surfaces by themselves. They may support procedural legitimacy, but they do not compute physics or establish biological, mechanical, medical, infrastructural, or cyber-physical safety.

Determinism boundary. Deterministic verification is valuable where the specification, state space, and transition relation support it, but it is neither universally available nor sufficient by itself. Safety-critical justification may legitimately combine formal proof, probabilistic risk analysis, empirical testing, simulation, safety cases, redundancy, bounded operating envelopes, monitoring, fail-safe behavior, and accountable residual-risk authorization. The required warrant strength is proportionate to consequence, uncertainty, reversibility, and domain practice. A deterministic proof of the wrong specification does not establish safety; a probabilistic warrant is not invalid merely because uncertainty remains.

Computational-source discipline. A material execution claim SHOULD distinguish the source that generated the candidate action from the source that warrants the action. If candidate generation and admissibility warrant are supplied by the same system, the claim record SHOULD state how circular self-confirmation is prevented and what independent check, validity-domain limit, or refusal condition applies.

Formal-conformance layer rule. A proof that an implementation satisfies a formal specification supports only the formal-conformance claim established by that proof. A physical or causal claim additionally requires a justified specification-to-reality mapping, declared validity domain, boundary conditions, uncertainty treatment, and domain evidence. Failure at the correspondence layer reopens every downstream claim that relied on the formal model as a physical warrant. **Institutionalized-uncertainty boundary.** A policy, risk acceptance, certification, or approval process that assigns responsibility without establishing the relevant operating envelope may govern who may test, monitor, stop, or review a system, but it MUST NOT be presented as reducing the underlying epistemic uncertainty unless new evidence actually does so.

If a physical-admissibility claim changes its model, validity domain, threshold, controller envelope, failure-mode account, or monitoring/shutoff assumptions, affected downstream gates must be rerun. Silent tuning may improve wording, but it does not preserve the original claim.

Capability Claim Integrity

When a material decision or public statement depends on a capability term such as agentic, autonomous, reasoning, understanding, self-improving, or intelligent, the record MUST state an operational definition and observable criteria before the term receives gate, authority, deployment, conformance, or communication weight. The strongest permitted claim is limited by the declared evidence surface, validity conditions, and evidence maturity stage.

A coordinated system MAY possess a bounded system-level operational capability even when the capability is implemented through ordinary components such as probabilistic inference, memory, planning, deterministic constraints, tool interfaces, environmental feedback, and human controls. Decomposing the system into those components does not by itself refute the system-level capability. At the same time, fluency, anthropomorphic language, market labels, task success, or user impression does not establish consciousness, personhood, intrinsic goals, moral agency, unrestricted autonomy, general intelligence, safety, lawful mandate, or ethical authority.

The record MUST test material alternative explanations and state a falsification or revision trigger. If the evidence cannot distinguish the claimed capability from a materially simpler explanation, the claim must be narrowed, marked assumption-bound or contested, restricted to exploratory use, or refused at the stronger level.

Claim-type examples

Claim type	Example	Minimum discipline
Empirical	"Injury rates decreased after the intervention."	Requires evidence contact, baseline, uncertainty, and alternative-explanation review.
Causal	"The policy caused the decrease."	Requires mechanism, counterfactual, intervention logic, or qualified causal warrant.
Formal	"CVaR was computed by the declared algorithm."	Requires definitions, assumptions, formula, inputs, and reproducible calculation surface.
Operational	"The system is operationally autonomous within the	Requires an operational definition, observable criteria,

Claim type	Example	Minimum discipline
	declared pilot envelope.”	system composition, evidence trace, alternatives, validity conditions, falsifier, explicit nonclaims, controls, and authority boundary.
Legal-regulatory	“This agency may lawfully authorize the trial.”	Requires current authority, jurisdiction, applicability analysis, interpretation status, and challenge or review path.
Normative	“This rights floor is adopted as non-compensatory.”	Requires declared ethical commitment, consistency check, rights compatibility, and challenge pathway.
Metaphysical horizon	“AIU language is used as a humility horizon.”	May guide meaning and restraint, but creates no Tier 1-3 evidence or gate authority.